

Ezra Stone's Section

Deliverable

The Uncorrelated Orbital Track Processing (UCTP) project to me presents an exciting opportunity to build upon my little experience with LLM technologies. My journey so far through Computer Science has had a large emphasis on data structures and algorithms. Instilling to me the critical need for efficient and reliable algorithms to process large scale data. The work on benchmarking UCTP algorithms as outlined in the original project plan is a testament to this. The ability to accurately classify and characterize space events is paramount for both civilian and military applications and I am driven by the idea of developing a solution that can improve these existing methods.

The core motivation for me lies in the idea of building a system that can improve itself. From my understanding the current methods being used are inefficient and can be greatly improved if done correctly. The concept of an agent which can autonomously generate, test, and tweak algorithms, represents to me the possibility of a profound shift. I am excited by the challenge of applying this groundbreaking AI system to solve a problem as complex and crucial as UCTP. It possesses an opportunity to move beyond the manual optimization of individual algorithms and explore an automated approach to discovery.

My interest in AI is a direct result of my prior projects and research that I've done in course work and my own free time. Through my own research I have seen the exponential growth of data in every aspect of technology. We've all probably witnessed how traditional methods struggle to keep pace with the sheer volume and velocity of information. This has made it very clear that the future of data analysis lies in more intelligent, autonomous systems. This is especially true in domains such as government, military, and space operations.

Work Distribution

From my personal point of view, the opportunity to learn and contribute to OpenEvolve is particularly exciting because of how new it is. The openEvolve framework being a key component of this project represents a new frontier in AI assisted code development. It's not just about using pre-trained models to generate code, it's about building a system that learns to “evolve” and improve its own solutions. This is a level of agentic intelligence that will move beyond automation and into the realm of co-creation. OpenEvolve’s modular design with distinct components for prompt sampling, LLM ensembles, and evaluation provide a flexible and powerful foundation for experimenting.

I am particularly interested in the evaluation process and how it can be used to provide a quantifiable fitness score for the evolving algorithms. The ability to objectively measure performance is what makes an evolutionary approach viable and I am excited to develop this component for our specific UCTP task. In addition to this the chance to work with the Gemini API and integrate it into a real-world project is an invaluable learning and project experience. I am confident that my skills and enthusiasm for this kind of work will allow me to make a significant contribution to this project's success.

Beyond the technical aspects of the project, I am also deeply motivated by the opportunity to work with the Air Force Research Laboratory (AFRL). The chance to contribute with the SDA TAP Lab and its mission to enhance space situational awareness is incredibly exciting. I am eager to collaborate with the team at the SDA TAP Lab and to gain a deeper understanding of the unique requirements and operational constraints of the space domain.

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